Monetary Policy and the Redistribution Channel

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Central Banking policies for Sustainability

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Main question

Is redistribution merely a side effect of monetary policy — or a transmission channel?

ggregation would not matter if we could be sure that the marginal propensities to spend from wealth were the same for creditors and for debtors. But the population is not distributed between debtors and creditors randomly. Debtors have borrowed for good reasons, most of which indicate a high marginal propensity to spend.

— Tobin (1982)

This Paper's Answer

- Redistribution IS a channel.

When winners from monetary expansions have higher MPCs than losers, redistribution amplifies aggregate consumption responses.



Three Redistribution Channels of Monetary Policy



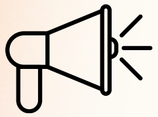
Earnings Heterogeneity

Monetary expansions i**ncrease labor and profit income.** The distribution of these gains is unequal — some agents benefit disproportionately



Fisher Channel

Unexpected inflation revalues nominal balance sheets. **Nominal creditors lose; nominal debtors gain.** Measured by **Net Nominal Positions** (NNP).



Interest Rate Exposure

Real rate **falls** create capital **gains/losses depending on balance-sheet duration.** Measured by **Unhedged Interest Rate Exposures (URE)**.

Individual Balance Sheets

Life-cycle model with perfect foresight, arbitrary maturity structure, separable preferences.

Intertemporal Budget Constraint

$$\Sigma\, q_t\, c_t = \Sigma\, q_t\, (y_t + w_t\, n_t) + \Sigma\, q_t\, [(-_1 b_t) + (-_1 B_t / P_t)]$$

Two Balance-Sheet Exposure Measures

NNP — Net Nominal Position

$\text{NNP} \equiv \Sigma\, Q_t\, (-_1 B_t / P_0)$

Present value of nominal assets. Measures exposure to unexpected inflation. Large NNP → loses from inflation.

URE — Unhedged Interest Rate Exposure

$\textbf{URE} \equiv y + wn + (-_1 B_0/P_0) + (-_1 b_0) - c$

Maturing assets minus liabilities (incl. consumption). Measures exposure to real rate changes.

Consumption Response to a Transitory Shock

For an optimizing agent under a transitory shock to income (dy), inflation (dP/P), and real rate (dR/R)

Formula –> dc = MPC × (dΩ + ψn·dw) – σ × c × MPS × dR/R:

Where the net wealth change dΩ decomposes as:

dy + n·dw
Income gain (human wealth effect)

– NNP · (dP/P)
Fisher effect: nominal asset revaluation

+ URE · (dR/R)
Rate effect: net asset /liability revaluation

Corollary 1

$MPC\hat{C} = MPC / (MPC + MPS) \geq MPC$

When borrowing-constrained or unable to access financial markets: MPS = 0, so $MPC\hat{C} = 1$. Consumption = pure wealth effect

MPC out of transitory income is the sufficient statistic for ALL balance-sheet revaluations — inflation, interest rates, and windfalls.

The Five Channels of Monetary Policy Transmission

$$dC = [\text{Aggregate Income}] + [\text{Earnings Heterogeneity}] - [\text{Fisher}] + ([\text{Int. Rate Exposure}] - [\text{Substitution}]) \cdot dR/R$$

Aggregate Income	• $\mathbb{E}[Y_i/Y \cdot \widehat{MPC}_i] \cdot dY$
Earnings Heterogeneity	• $\text{cov}(\widehat{MPC}_i, dY_i - Y_i \cdot dY/Y)$
Fisher Channel	• $-\text{cov}(\widehat{MPC}_i, NNP_i) \cdot dP/P$
Interest Rate Exposure	• $\text{cov}(\widehat{MPC}_i, URE_i) \cdot dR/R$
Substitution Channel	• $-\mathbb{E}[\sigma_i(1-\widehat{MPC}_i)c_i] \cdot dR/R$

Representative Agent vs. Heterogeneous Agent

Representative Agent World	Heterogeneous Agent World
All covariance terms = 0 **$dC = MP\hat{C} \cdot dY - \sigma(1-MP\hat{C}) \cdot C \cdot (dR/R)$** **Only aggregate income effect and intertemporal substitution matter.** **Policy works only through the interest rate path.** **Tipically low $MP\hat{C}$ → substitution dominates.**	Three additional redistribution channels active. Redistribution AMPLIFIES if: • $cov(MP\hat{C}_i, URE_i) < 0$ • $cov(MP\hat{C}_i, NNP_i) < 0$ • $cov(MP\hat{C}_i, Y_i) < 0$ Borrowers/low-income agents have higher MPCs than savers → money flows to those who spend more. Monetary policy is more powerful than RA predicts.

The Aggregate Income Channel

Aggregate Income = $\dot{Y}/Y \cdot MPC\hat{} \mathcal{M}$

1

What does M measure?

✦ It is the average income-weighted MPC

✦ Quantify how much of the additional income is spent on consumption

✦ With representative agent: M = MPĈ

Effect of rate cut: must reinvest at lower rates → real wealth loss

→ These agents LOSE from ↓R

2

Keynesian multiplier

•In general equilibrium (dC = dY), the multiplier is:

1 / (1 – MPĈ)

Amplifies the first impact of intertemporal substitution. Channel (1) is therefore also the vector of the multiplicative effects of the other channels

Interaction with the income heterogeneity channel: The distribution of dY earnings between agents with different MPCs determines whether the aggregate effect amplifies or attenuates the expansion. If low-income workers (high MPC) benefit more, the effect is amplified.

Earnings Heterogeneity

Earnings Heterogeneity Channel = $\gamma \cdot \text{cov}(\widehat{MPC}_i, Y_i/Y) \cdot dY$

1

Empirical Basis (cov < 0)
Low-income agents have higher MPCs — consistent across ALL three datasets in the paper and much of the prior literature

2

Cyclicality of Income Risk (γ < 0)
income risk is countercyclical — low earners benefit more in booms.

3

Amplification Condition

If $\gamma < 0$ and $\text{cov}(\widehat{MPC}, Y) < 0 \rightarrow$ their product is positive → channel amplifies the aggregate income effect of expansion.

4

Direct Evidence

Coibion et al. (2017): identified monetary policy shocks reduce income inequality in the Consumer Expenditure Survey.

The Fisher Channel

Fisher Channel = $-\text{cov}(\widehat{MPC}_i, NNP_i) \cdot dP/P$

1

Who has positive NNP?
Rich, older households (bonds, deposits). Lose real wealth when inflation rises unexpectedly.

2

Who has negative NNP?
Young middle class with mortgage debt. Gain from inflation (real debt burden falls).

3

Amplification Logic

If nominal borrowers have higher MPCs (cov < 0), surprise inflation redistributes to higher spenders → aggregate consumption rises.

4

Empirical Magnitude

Covariance $\widehat{MPC}$–NNP is negative in all three surveys, but the elasticity is small: a 1% surprise inflation raises C by at most ~0.1%.

The Interest Rate Exposure Channel

Interest Rate Exposure Channel = $\text{cov}(\hat{MPC}_i, URE_i) \cdot dR/R$

1

Who has URE > 0? (Savers)

• Agents with large savings in short-term CDs or deposits

• Savers who reinvest maturing bonds

• Those with income > consumption at present

Effect of rate cut: must reinvest at lower rates → real wealth loss

→ These agents LOSE from ↓R

2

Who has URE < 0? (Borrowers)

• ARM mortgage holders (adjustable rate)

• Short-term borrowers rolling over debt

• Consumption plan exceeds current income

Effect of rate cut: debt service costs fall → real wealth gain

→ These agents GAIN from ↓R

The Intertemporal Substitution

Intertemporal Substitution Channel = $\sigma (1 - MP\hat{C}) \cdot c\, dR/R$

1

Parameters

•σ — Intertemporal substitution elasticity (EIS)

$MP\hat{C}$ — marginal propensity to consume total

MPS = 1 - $MP\hat{C}$ — marginal propensity to save

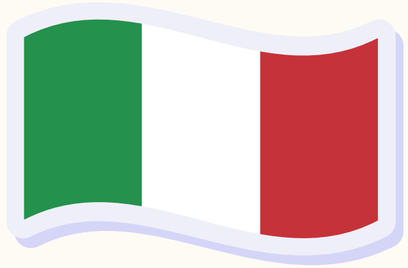
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Central role of the σ

If σ is small (0.1–0.5, as macroeconomists estimate), the substitution effect is weak. The redistributive IRE channel thus becomes comparable in magnitude to the substitution channel itself.

The only channel also present in the standard representative agent model. A reduction in the real rate lowers the relative cost of consumption today compared to tomorrow: agents replace future consumption with present consumption.

Three Surveys, Three Identification Methods



SHIW

Italian Survey of Household Income & Wealth

Self-reported MPC

Households asked what share of a hypothetical windfall they would spend immediately (Jappelli & Pistaferri 2014). Provides individual-level MPC — ideal for covariance computation.



PSID

Panel Study of Income Dynamics

Semi-structural (BPP)

Blundell, Pistaferri & Preston (2008): uses income process and consumption function restrictions to recover group-level MPC from transitory shocks.



CE

Expenditure Survey

Tax Rebate IV

Johnson, Parker & Souleles (2006): randomized timing of 2001 tax rebate receipt identifies MPC out of an anticipated income transfer.

Constructing URE and NNP from Survey Data

URE = Y − T − C + A − L

Y = Gross Income	Labor, dividend, interest income + realized capital gains (from all sources)
T = Taxes net of Transfers	All taxes minus all transfers; Y − T = disposable income
C = Consumption	Nondurables + (1−ε)·durables + rents + interest payments + amortization
A = **Maturing Assets**	Deposits (2-quarter duration benchmark), bonds (4-year US / 7-year Italy)
L = Maturing Liabilities	ARM payments (3-quarter duration benchmark), credit cards (2-quarter)

NNP = Nominal Assets(deposits + bonds) − Nominal Liabilities(mortgages+consumer credit)

Summary Statistics from the Three Surveys

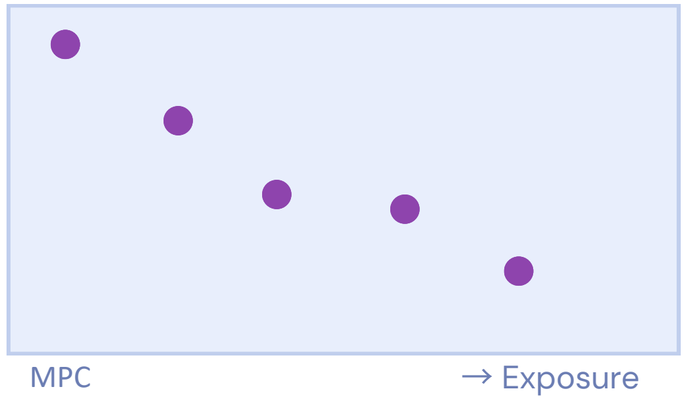
All variables normalized by average consumption in each survey (facilitates comparison of elasticities)

Variable	SHIW Mean	SHIW SD	PSID Mean	PSID SD	CE Mean	CE SD
Net income (Y−T)	**119**	**83**	**142**	**164**	**114**	**101**
Consumption (C)	**100**	**94**	**100**	**258**	**100**	**90**
Maturing assets (A)	93	248	130	506	53	177
Maturing liabilities (L)	31	140	51	144	52	147
URE	81	304	121	606	18	243
Net nominal position (NNP)	24	264	−0.55	443	−0.68	247
Gross income (Y)	127	104	169	251	123	109
MPC (mean)	47	35	—	—	—	—

Main Empirical Result: MPC Declines with URE, NNP, and Income

MPC vs URE

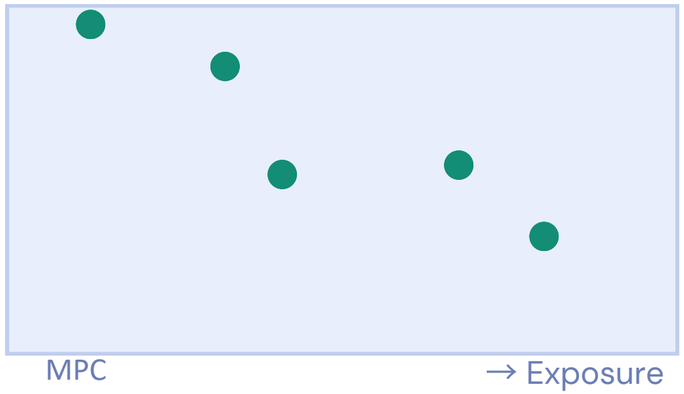
(Interest Rate Exposure Channel)



Negative correlation in all 3 datasets. Higher URE → lower MPC. Borrowers (URE < 0) spend more at the margin than savers (URE > 0).

MPC vs NNP

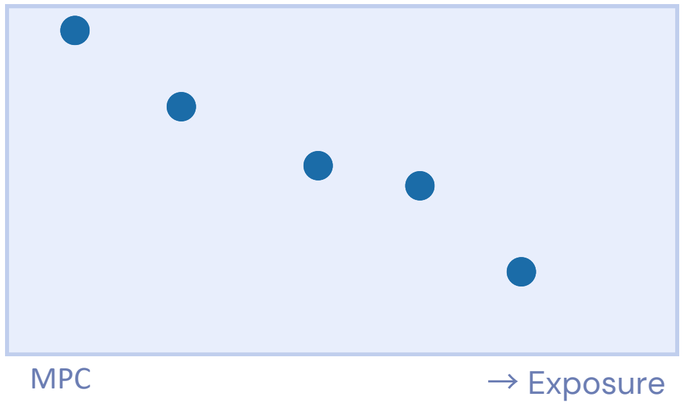
(Fisher Channel)



Overall negative correlation. Net nominal debtors have higher MPCs. Slight hump in center (wealthy hand-to-mouth). The patterns are weaker but consistent.

MPC vs Income

(Earnings Heterogeneity Channel)



Strong negative correlation everywhere. Low-income households have dramatically higher MPCs, consistent with vast prior literature.

Figure 2 :consistent pattern across all three surveys and all three exposure measures

Role of Asset Durations and Structural Insights

The magnitude of the interest rate exposure channel depends critically on financial market structure

Survey	Quarterly	Short	Benchmark	Long	Annual
SHIW	**−0.16***	**−0.20***	**−0.10***	**−0.07***	**−0.06***
PSID	**−0.14***	**−0.13***	**−0.12***	**−0.11***	**−0.11***
CE	−0.55	−0.48	−0.23	−0.22	−0.23

Asymmetry: rate hikes have larger consumption effects than cuts (borrowers constrained). Alternative to downward wage rigidity explanation.

Shorter durations → $\mathscr{E}$R more negative: with short-term assets, rate cuts generate less capital gains for low-MPC savers, so redistribution is larger.

Country comparison: consumption reacts much more strongly to monetary shocks where ARM mortgages dominate instead of fixed-rate.

$\mathscr{E}$R by Duration Scenario

Conclusions

- Redistribution IS a channel of monetary policy transmission – not just a side effect.
- Three channels: Earnings Heterogeneity, Fisher (NNP), Interest Rate Exposure (URE).
- All three channels likely **AMPLIFY** monetary expansion – confirmed empirically in Italian and US data.
- The MPC out of transitory income is the key sufficient statistic for all balance-sheet revaluations.